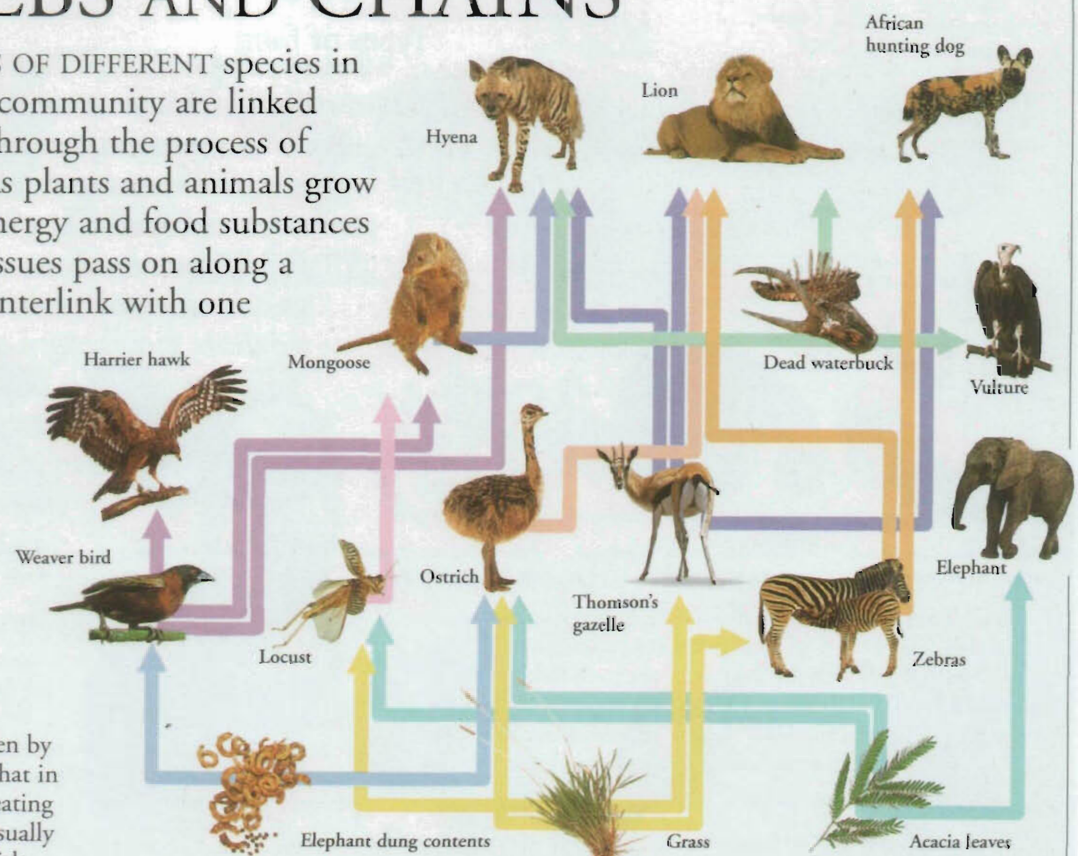
FOOD WEBS AND CHAINS

THE LIVES OF DIFFERENT species in a wildlife community are linked together through the process of feeding. As plants and animals grow and are eaten by others, energy and food substances locked up in their body tissues pass on along a chain. These food chains interlink with one another, and the resulting network is called a food web. The number of different animals and plants in a community is naturally balanced. If the balance is upset it can affect the whole web.



Food webs

In a simple food chain, a plant is eaten by a herbivore (a plant-eating animal), that in turn is eaten by a carnivore (a meat-eating animal). In nature, food chains are usually longer than this, and they connect with other chains to form a web. The arrows in this diagram show how different plant foods on a typical African savannah are eaten by a range of animals, that, in turn, provide food for various other animals.

Decomposers
Some animals, fungi, and bacteria feed on dead or waste plant and animal tissue. They turn it back into simple substances, which plants use to grow.

Producers
In ecology, plants are called producers. They start the food chain by using the sun's energy to produce food from simple substances.

Consumers
Animals are known as consumers because they get the biological material they need for life from the plants or other animals that they eat or consume.

Top predator
The tawny owl at the top of this food chain is known as the top predator. An owl needs to eat many weasels and rodents to meet its energy needs.

Trophic pyramids

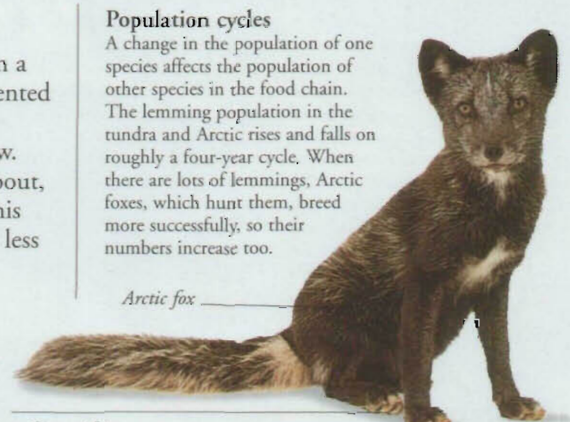
Ecologists call each stage in a food chain a trophic level. These levels can be represented as a pyramid. Animals use much of the energy they gain from their food to grow. They also use energy to live, to move about, breed, feed, and avoid their enemies. This means that at each trophic level there is less energy available to the next level.

Secondary consumers
Weasels are secondary consumers because they get energy from the plants through other consumers. There are more weasels in a community than the owls that hunt them.

Primary consumers
Mice and voles get energy directly from plants. They use a lot of energy; many are needed to support the weasels.

Primary producers
Energy is stored in plants. It takes a large quantity to support the rodents.

Population cycles
A change in the population of one species affects the population of other species in the food chain. The lemming population in the tundra and Arctic rises and falls on roughly a four-year cycle. When there are lots of lemmings, Arctic foxes, which hunt them, breed more successfully, so their numbers increase too.



Upsetting the balance
When the giant African land snail was taken to Pacific islands, the snails destroyed vegetation because there was nothing to prey on them. Another type of snail was released to eat their eggs, but these began to wipe out the native *Partula* snail instead.

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